Siliconpolitik v2025

Challenges and Opportunities for India

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The Big Picture

The focus going ahead should be on building *asymmetric leverage* by capitalising on our strengths

Approved and Under Construction Projects (as of Sep 2025)

- 1 CMOS Chip Fabrication plant (<10% of India's demand)
- 1 Compound Semiconductor fabrication and packaging plant
- 8 Assembly, Test and Packaging plants

Fabless Firms

- Chip design projects from 23 companies approved under P-DLI scheme with a total project cost of $\sim$ 700 crore.
- goal was to support 100 start-ups in their go-to-market strategy by Dec 2026.

Overall Disbursements

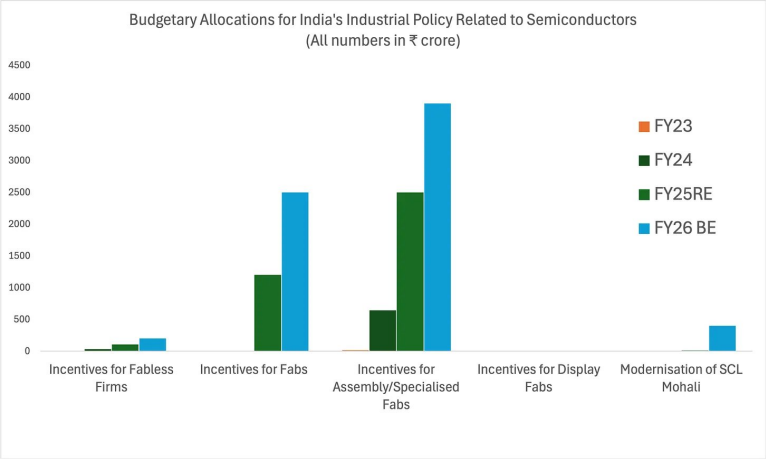

Figure 1: Source: Compiled from budget data

So what?

- It's a long journey, we have only taken the first few steps
- Impressive financial support given India's other challenges
- Need to execute each of the projects well
- India won't become *atmanirbhar* in semiconductors (neither is Taiwan - $60bn imports)
- Need to seek leverage by developing some asymmetric leverage in key segments

Likely: Mature Nodes Overcapacity

- 107 new fabs are set to come up by 2030; 91 already work in progress
- Mature nodes segment highly likely to face overcapacity and 'China Shock'
- Demand for advanced fab nodes likely to sustain
- Driven by national security concerns not market demand
- Implication: mature node fabs need anchor customers like MoD. Trusted Foundry Accreditation will help.

Blackswan: A Taiwan Blockade or Invasion will rock the Industry

Figure 2: source: Gartner and Crédit Agricole S.A.

Blackswan: A Taiwan Blockade or Invasion will rock the Semiconductor Industry

- TSMC and UMC have significant presence in advanced *and* mature nodes
- Invasion or blockade will lead to shortages for at least two-three years
- Implication 1: A flexible sovereign fund to support Indian companies and start-ups in case of this eventuality
- Implication 2: Aggregate demand from Indian players and encourage them to work with foundries outside Taiwan

Highly Likely: Global Talent Shortage

- global shortage of *fab process engineers* and *experienced technicians* is the single biggest bottleneck to all national chip ambitions.
- Implication: need focus on talent development in these segments

Likely: More Restrictions and Controls

Strategic Objective	Instruments Used	Underrated Repercussions
Denial	Sanctions and controls	Difficult to sustain; incentives for backroom deals with adversary
	Restricting movements of semiconductor labour	Can slow down technical progress
	End-use restrictions	Encourages adversary to build local capacity in a focused manner
	Investment screening	Can slow down technical progress
Outpace adversary	industrial espionage to steal secrets, targeted poaching	Invites stricter controls on professionals from the stealing country
	Build partnerships	Self-sufficiency is a myth
	Indigenisation and industrial policy	Difficult to sustain
	Economic coercion	Self-damage
	Encouraging the movement of labour	Human capital is the dark horse in the race for semiconductor supremacy
	Increase dependence and control	Helps manage the adversary's pace to an extent
Increase supply chain resilience	Promote open-source hardware	Still a nascent field
	Mapping supply chains	Firms care more about resilience than governments
	Build partnerships to create a resilient supply chain.	Self-sufficiency is a myth.
Influence minds and actions	Hardware espionage	Limited impact on national power

Figure 3: Source: Author

Packaging Becomes Cool

- The game is shifting from the "Fab" (Front-End) to the "OSAT" (Back-End).
- "System-in-Package" (SiP) allows mix-and-match of small chips ("chiplets") from different fabs (e.g., 3nm AI + 22nm I/O + 65nm Power).
- Advanced Packaging is now strategically important
- Implication: Develop OSAT as leverage, not just for import substitution

Master the Niche

- For defence, Compound Semiconductors like GaN are more important than cutting-edge silicon.
- Gallium Nitride (GaN) is the material for AESA Radars and Electronic Warfare. We are not that far from the frontier in this segment.
- Implications:
  - DRDO can license its PDK to select Indian players
  - It takes $3-4 Mn and 3 years to bring a "qualified, mil-grade" GaN chip. Such quantum of funding not available.
  - Long-term contracts, trusted foundry accreditation, buyer of first resort

Unlocking India's Strengths in the Fabless Segment

- Bring the DLI programme into the ISM
- Fund challenge grants for nationally important projects
- have good IP clear and good IP ownership rules for startups
- Bet on firms that are building out of India even if not Indian-owned
- Embrace open architectures like RISC-V to reduce dependence on one company or country
- Modernise and open SCL for prototyping. Open PDKs to Indian firms

Chip Atmashakti not Chip Atmanirbharta

- Sovereignty as self-sufficiency is impractical
- Sovereignty as building baseline capabilities is ongoing
- Sovereignty through research, education, and innovation policy is the long-term solution
- Sovereignty as having asymmetric technological power is the strategic necessity